Graphs and Transformations

Chapter 2 · Original slide 26 · 4: Quadratic Graphs

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

y

A 'sketch' in maths has a clearly defined meaning:

• We care about the general shape of the graph, not exact points.

• The axes should have no scale.

• Generally the only coordinates indicated are intercepts with the axis or other points of interest (e.g. intersections of multiple graphs)

Features needed in sketch of quadratic?

Recall a root of a function is where the output, in this case the y value, is 0.

Roots

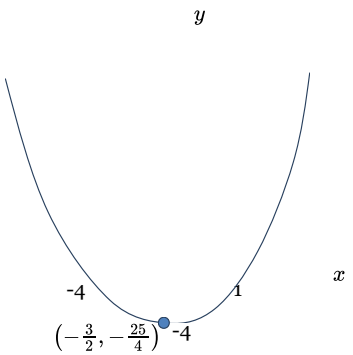
x

General shape: Smiley face or hill?

y-intercept

Turning point

Chapter 2 · Original slide 27 · Example



Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

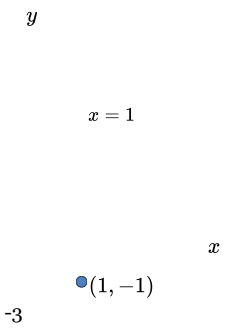
y-intercept:

Turning point:

Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

Graphs and Transformations

Chapter 2 · Original slide 28 · Example



Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

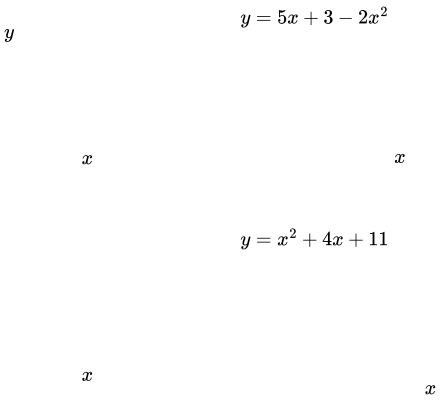
Roots:

y-intercept:

Turning point:

Line of reflection:

Chapter 2 · Original slide 29 · Test Your Understanding



Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

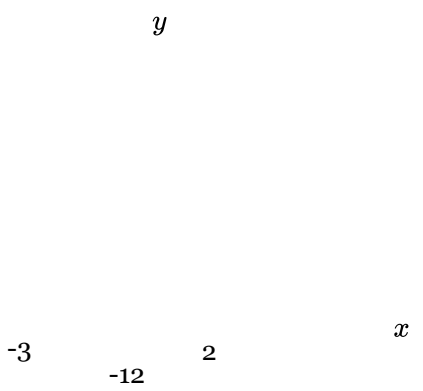
a c

$y = x^2 + 4$

b d

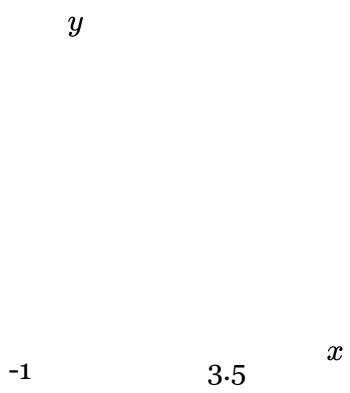
$y = x^2 - 7x + 10$

Chapter 2 · Original slide 30 · Determining the Equation using a Graph



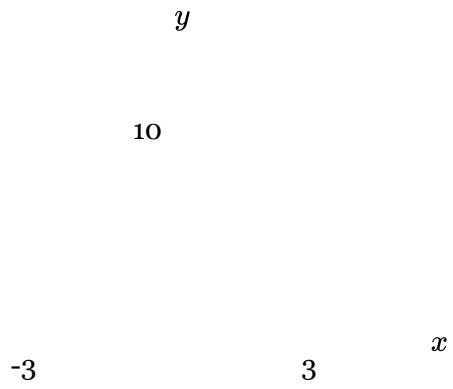
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

Chapter 2 · Original slide 31 · Further Example



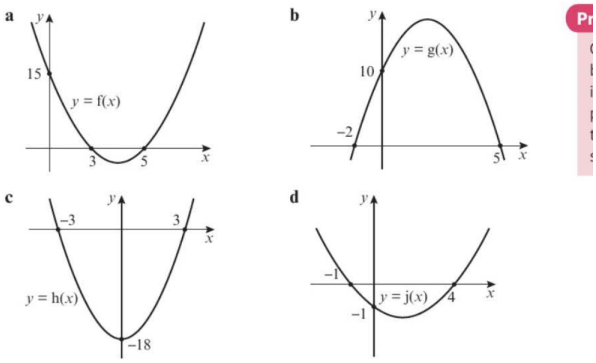
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

Chapter 2 · Original slide 32 · Test Your Understanding



Determine an equation of this quadratic graph.

Chapter 2 · Original slide 33 · Exercise 2F Pure Y1 Page 30



2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a,b,c for each function.

Chapter 2 · Original slide 34 · Exercise 2F Pure Y1 Page 30

3 The graph of $y = ax^2 + bx + c$ has a minimum at (5,-3) and passes through (4,0). Find the values of a,b,c.

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

x

Extension Question:

y

Chapter 4 · Original slide 4 · Polynomial Graphs

In Chapter 2 we briefly saw that a polynomial expression is of the form:

$a + bx + cx^2 + dx^3 + ex^3 + \ldots$

where $a, b, c, d, e, \ldots$ are constants (which could be 0).

The order of a polynomial is its highest power.

Order	Name
0	
1	
2	
3	
4	
5	

These are covered in Chapter 5.

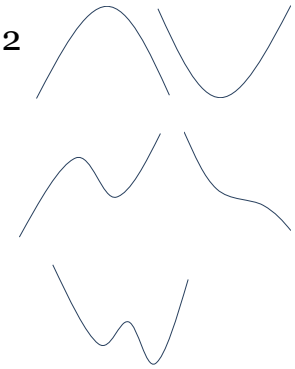
We will cover these now.

Chapter 2 explored the graphs for these.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

Chapter 4 · Original slide 5 · Polynomial Graphs

2



What property connects the order of the polynomial and the shape?

Order:

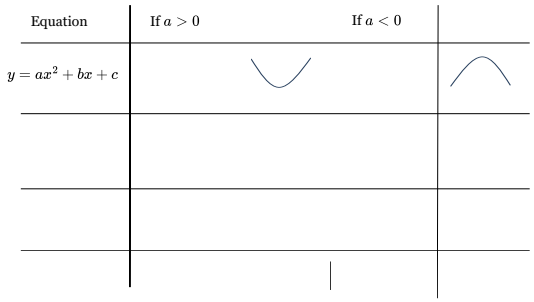
Bro Note: ...Actually this is not strictly true, e.g. consider $y = x^4$, which has a U shape. But this is because multiple turns are being squashed into a single point.

In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

3

4

Chapter 4 · Original slide 6 · Polynomial Graphs



e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Resulting Shape

As $x \to \infty, y \to \infty$
As $x \to -\infty, y \to \infty$

$y = ax^3 + bx^2$
$+cx + d$

$y = ax^5 + bx^4 + \ldots$

(And we have the opposite if $a < 0$)

Resulting Shape

As $x \to \infty, y \to -\infty$
As $x \to -\infty, y \to -\infty$

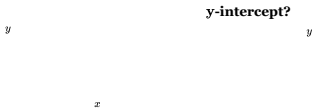
$y = ax^4 + bx^3$
$+cx^2 + dx + e$

If a>0, what therefore can we say about the shape if:

• The order is odd:

• The order is even:

Chapter 4 · Original slide 7 · Cubics





Sketch the curve with equation
$y = x^2(x-1)$

Shape?

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = (x-2)(1-x)(1+x)$

Features you must consider:

Tip: No need to expand out the whole thing. Just mentally consider the x terms multiplied together.

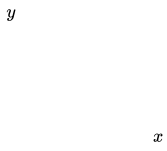
Roots?

This is sort of because the curve crosses at 0 then immediately crosses at 0 again!

Tip: It's incredibly easy to forget to write in one of the intercepts. So don't!

Chapter 4 · Original slide 8 · Cubics





Sketch the curve with equation
$y = (x-4)^3$

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = (2-x)(x+1)^2$

Shape?

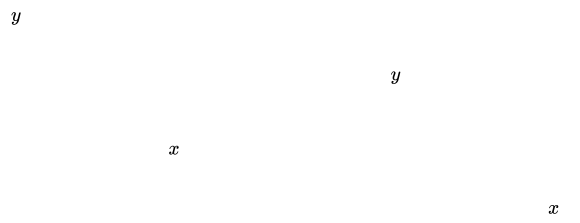
Roots?

y-intercept?

Exam Notes: The term 'point of inflection' has been removed from the new A Level syllabus (bad idea!!).

A point of inflection is where the curve goes from 'convex' to 'concave' (or vice versa), i.e. curves in one direction before and curves in another direction after. You might have encountered these terms in Physics.

Chapter 4 · Original slide 9 · Cubics with Limited Roots



Sketch the curve with equation

$y = (x + 1)\left(x^2 + x + 1\right)$

Shape?

Roots?

y-intercept?

However, in Chapter 12, we'll be able to work turning points using 'differentiation', and hence conclude that it doesn't have any!

Chapter 4 · Original slide 10 · Finding the equation yourself

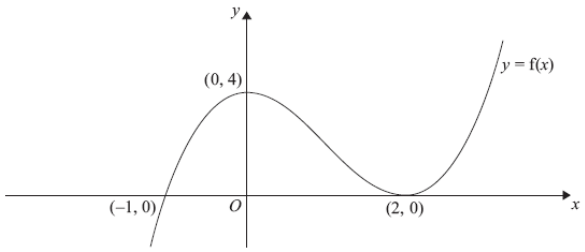


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.

The curve C passes through the point (–1, 0) and touches the x-axis at the point (2, 0).

The curve C has a maximum at the point (0, 4).

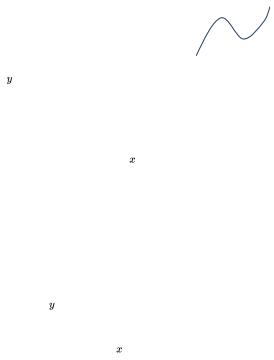
The equation of the curve C can be written in the form.

$y = x^3 + ax^2 + bx + c$

where a, b and c are integers.

(a) Calculate the values of a, b, c.

Chapter 4 · Original slide 11 · Test Your Understanding



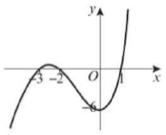
1
Sketch the curve with equation
$y = x(x-3)^2$

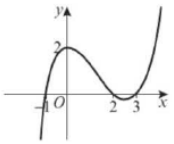
2
Sketch the curve with equation
$y = -(x+2)^3$

3
A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

N
Sketch the curve with equation
$y = 2x^2(x-1)(x+1)^3$

Chapter 4 · Original slide 12 · Exercise 4A –Y1 Pure pages 63





5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b,c,d are real constants.

a Find the values of b, c, d. (3 marks)
b Write down the coordinates of the point where the curve crosses the y-axis. (1 mark)

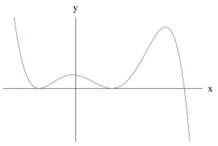
Graphs and Transformations

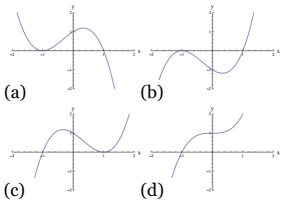
Chapter 4 · Original slide 13 · Exercise 4A –Y1 Pure pages 63

7 Given that $\mathrm{f}(x) = (x-10)(x^2-2x)+12x$

a Express $\mathrm{f}(x)$ in the form $x(ax^2+bx+c)$ where a, b, c are real constants.
b Hence factorise $\mathrm{f}(x)$ completely.
c Sketch the graph of $y = \mathrm{f}(x)$ showing clearly the points where the graph intersects the axes.

Chapter 4 · Original slide 14 · Exercise 4A Extension





Extension

2

[MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?

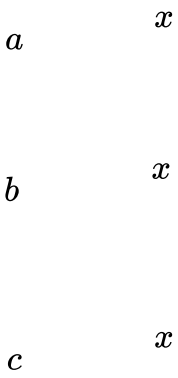
A) $y = (3-x)^2(3+x)^2(1-x)$
B) $y = -x^2(x-9)(x^2-3)$
C) $y = (x-6)(x-2)^2(x+2)^2$
D) $y = (x^2-1)^2(3-x)$

1

[MAT 2012 1E] Which one of the following equations could possibly have the graph given below?

Cubics can sometimes be factorised by pairing the terms:

Chapter 4 · Original slide 15 · Recap



If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$:

Chapter 4 · Original slide 16 · Quartics



If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.

Recall that if the x^4 term is positive, the 'tails' both go upwards, otherwise downwards.

Sketch the curve with equation

$y = x(x+1)(x-2)(x-3)$

Shape:

Roots:

y-intercept:

Sketch the curve with equation

$y = (x-2)^2(x+1)(3-x)$

Shape:

Roots:

y-intercept:

Chapter 4 · Original slide 17 · Quartics



Sketch the curve with equation
$y = (x+1)(x-1)^3$

Sketch the curve with equation
$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a ‘touch’ point.

-1 root only appears once so line crosses at $x = -1$

+1 root triple repeated so point of inflection at $x = 1$

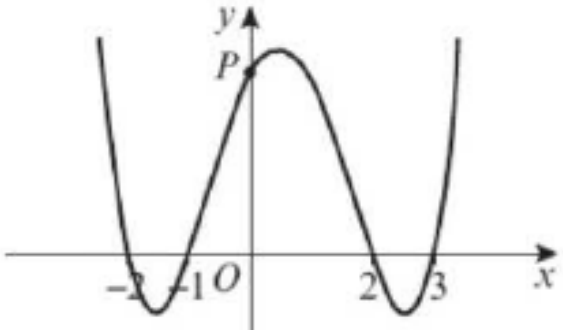
Chapter 4 · Original slide 18 · Test Your Understanding



Sketch the curve with equation
$y = x^2(x+1)(x-1)$

Sketch the curve with equation
$y = -(x+1)(x-3)^3$

Chapter 4 · Original slide 19 · Exercise 4B – Pure Y1 Pages 65-66



3 The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b,c,d,e are real constants.

a Find the coordinates of point P. (2 marks)

b Find the values of b, c, d, e. (3 marks)

4 Sketch the graph of
$y = (x + 5)(x - 4)(x^2 + 5x + 14)$.

Chapter 4 · Original slide 20 · Exercise 4B Extension

Extension

1

[STEP I 2012 Q2a]

a. Sketch $y = x^4 - 6x^2 + 9$

b. For what values of b does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

By changing b, we shift the graph up and down.
Then we can see that:

i) 0 roots:

ii) 1 root:

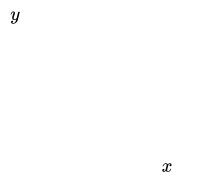
iii) 2 roots:

iv) 3 roots:

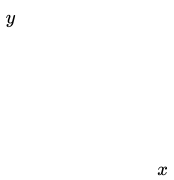
v) 4 roots:

b)

Chapter 4 · Original slide 21 · GCSE RECAP: Reciprocal Graphs



Sketch $y = \frac{1}{x}$



Sketch $y = -\frac{3}{x}$

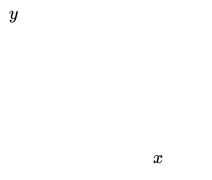
Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

Notice the distance between this line and the x-axis (i.e. the line $y = 0$) gradually decreases as the lines go off towards infinity. The line $y = 0$ is known as an asymptote of the graph.

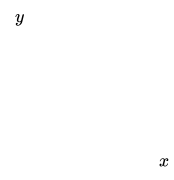
! An asymptote is a line which the graph approaches but never reaches.

Asymptotes of $y = \frac{a}{x}$:

Chapter 4 · Original slide 22 · Reciprocal Graphs



This is new to the A Level 2017 syllabus.



Sketch $y = \frac{3}{x^2}$

Sketch $y = -\frac{4}{x^2}$

Hint: Note that anything squared will always be at least 0.

Chapter 4 · Original slide 23 · Reciprocal Graphs

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

y

$y = \frac{3}{x}$

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

$y = \frac{1}{x}$

x

Chapter 4 · Original slide 24 · Exercise 4C – Pure Y1 page 67

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$
d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Chapter 4 · Original slide 25 · Exercise 4C – Pure Y1 page 67

2 Use a separate diagram to sketch each pair of graphs.
a $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$

Chapter 4 · Original slide 34 · Transformations of Functions

Suppose $f(x) = x^2$



Sketch $y = f(x)$:

Sketch $y = f(x + 2)$

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x + 2)$?

Graphs and Transformations

Chapter 4 · Original slide 35 · Transformations of Functions

This is all you need to remember when considering how transforming your function transforms your graph...

Affects which axis?

What we expect or opposite?

Change inside $f()$

Change outside $f()$

Therefore...

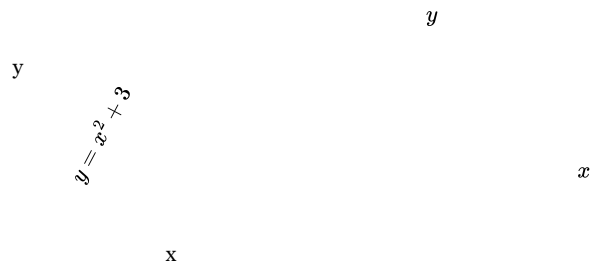
$y = f(x - 3)$

$y = f(x) + 4$

$y = f(5x)$

$y = 2f(x)$

Chapter 4 · Original slide 36 · Sketching transformed graphs



Sketch $y = \frac{2}{x+1}$

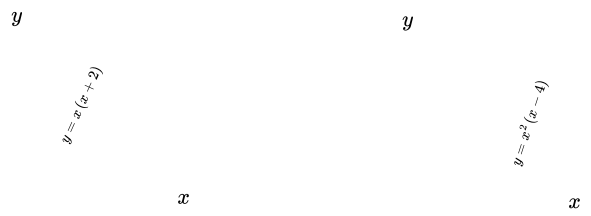
Sketch $y = x^2 + 3$

Imagine a sketch of $y = x^2$ and then do the translation, ensuring you adjust any intercepts with the axes.

The transformation might result in new intercepts or roots. You can find these in the usual way. Do not forget them!

Draw asymptotes using a dotted line and write its equation on it.

Chapter 4 · Original slide 37 · More Examples



Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$y = (2x)^2(2x-4)$

$y = (x-a)(x-a+2)$

$y = (2x)^2(2x-4)$

Note that our intercepts are in terms of a.

Chapter 4 · Original slide 38 · Reflections of Graphs

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

You did this at GCSE. The minus is outside the function, so affects the output, i.e. the y value.

The y values are negated, resulting in a reflection in the x-axis.

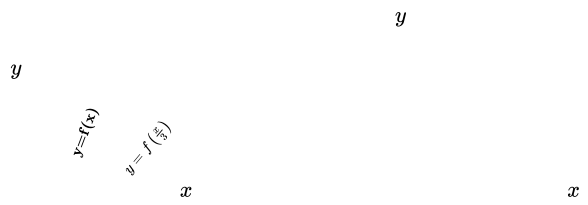
y

$y = f(x)$

$\mathbf{y = -f(x)}$

x

Chapter 4 · Original slide 39 · Test Your Understanding



Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.

If $y = (x + 1)(x - 2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.

Chapter 4 · Original slide 40 · Exercise 4E – Pure Y1 page 74-75 (translations)

7 The point P(4,-1) lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x - 2)$.
b State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

Chapter 4 · Original slide 41 · Exercise 4E – Pure Y1 page 74-75 (translations)

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$.
b The point with coordinates (-1,0) lies on the curve with equation $y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find the two possible values of a.

Chapter 4 · Original slide 42 · Exercise 4E – Pure Y1 page 74-75 (translations)

12 a Sketch the graph of $y = x(x + 1)(x + 3)^2$.
b Find the possible values of b such that the point (2,0) lies on the curve with equation $y = (x + b)(x + b + 1)(x + b + 3)^2$.

Chapter 4 · Original slide 43 · Exercise 4F – Pure 1 Page 78 (stretches/reflections)

7 The point P(2,-3) lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(2x)$.
b State the coordinates that point P is transformed to on the curve with equation $y = 4\mathrm{f}(x)$.

8 The point Q(-2,8) lies on the curve with equation $y = \mathrm{f}(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = \mathrm{f}(\frac{1}{2}x)$.

Chapter 4 · Original slide 44 · Exercise 4F – Pure 1 Page 78 (stretches/reflections)

9 a Sketch the graph of $y = (x-2)(x-3)^2$.
b The graph of $y = (ax-2)(ax-3)^2$ passes through the point (1,0). Find two possible values for a.

Chapter 4 · Original slide 45 · Effect of transformation on specific points

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.

Where would each of these points end up?

$\mathbf{y=}f(x)$	$(\mathbf{4},\mathbf{3})$	$(\mathbf{1},\mathbf{0})$	$(\mathbf{6},\mathbf{-4})$
$y=f(x+1)$			
$y=f(2x)$			
$y=3f(x)$			
$y=f(x)-1$			
$y=f\left(\frac{x}{4}\right)$			
$y=f(-x)$			
$y=-f(x)$			

Chapter 4 · Original slide 46 · Test Your Understanding

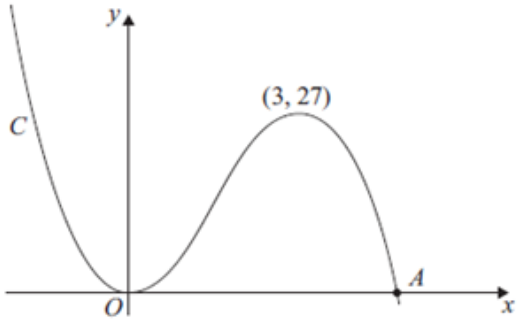


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point (3,27) and C cuts the x-axis at the point A.

(a) Write down the coordinates of the point A. (1)

(b) On separate diagrams sketch the curve with equation

(i) $y = \mathrm{f}(x + 3)$,

(ii) $y = \mathrm{f}(3x)$.

On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curves cross or meet the coordinate axes. (6)

The curve with equation $y = \mathrm{f}(x) + k$, where k is a constant, has a maximum point at $(3, 10)$.

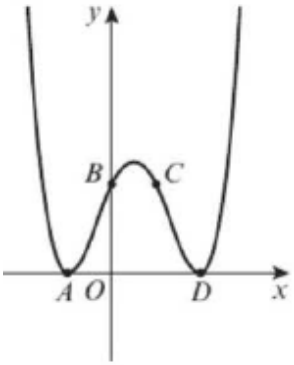
(c) Write down the value of k. (1)

Chapter 4 · Original slide 47 · Exercise 4G Pure Y1 Pages 80

5 The point P(2,1) lies on the graph with equation $y = \mathrm{f}(x)$.

a On the graph of $y = \mathrm{f}(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.
b Write down the coordinates of the point to which P maps under each transformation
i $\mathrm{f}(x - 4)$ **ii** $3\mathrm{f}(x)$ **iii** $\frac{1}{2}\mathrm{f}(x) - 4$

Chapter 4 · Original slide 48 · Exercise 4G Pure Y1 Pages 80



6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points A(-1,0),B(0,2),C(1,2),D(2,0) lie on the curve. Sketch the following graphs and give the coordinates of the points A,B,C,D after each transformation:

a $y + 2 = \mathrm{f}(x)$ **b** $\frac{1}{2}y = \mathrm{f}(x)$ **c** $y - 3 = \mathrm{f}(x)$ **d** $3y = \mathrm{f}(x)$

e $2y - 1 = \mathrm{f}(x)$

Chapter 5 · Original slide 4 · Let's start simple…What is the equation of this line?And more importantly, why is it that?



Let's start simple...

What is the equation of this line?

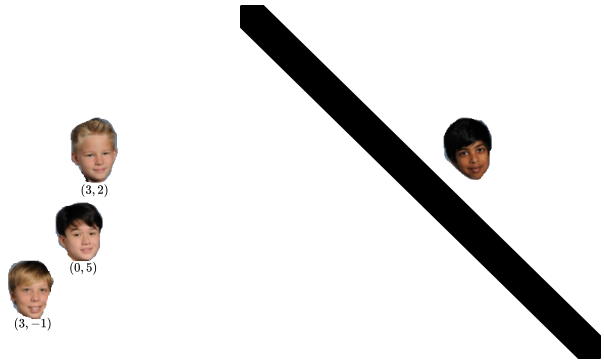
And more importantly, why is it that?

4
3
2
1
-1
-2
-3
-4

x -5 -4 -3 -2 -1 0 1 2 3 4 5 6

For any point we pick on the line, the x value is always 2.

Chapter 5 · Original slide 5 · Lines and Equations of Lines



! A line consists of all points which satisfy some equation in terms of x and/or y.

Anirudh's Line Club

Membership Rules:
$2x + y = 5$

Chapter 5 · Original slide 6 · Examples

This means we can substitute the values of a coordinate into our equation whenever we know the point lies on the line.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Chapter 5 · Original slide 7 · Test Your Understanding

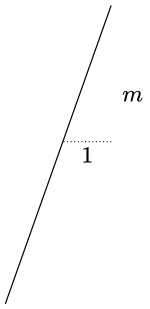
Determine where the line $x + 2y = 3$ crosses the:

a) y-axis:

b) x-axis:

What mistakes do you think it's easy to make?

Chapter 5 · Original slide 8 · Recap of gradient



The steepness of a line is known as the gradient.

It tells us what y changes by as x increases by 1.

So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?

How would that give us a suitable formula for the gradient m?

$m = \frac{\Delta y}{\Delta x}$

Δ is the (capital) Greek letter “delta” and means “change in”.

Textbook Note:

The textbook uses $m = \frac{y_2 - y_1}{x_2 - x_1}$ for two points (x_1, y_1) and (x_2, y_2).
Reasons I don’t use it for non-algebraic coordinates:

• Students often get the y_1 and y_2 the wrong way round (or with the x’s)

• Students often make sign errors when dealing with negatives, e.g. $(-3) - (-4)$

• It can’t be done as easily mentally,

• Students see it as “yet another formula to learn” when really all you need is to appreciate is what gradient is, i.e. “y change per x change”.

Chapter 5 · Original slide 9 · Examples

Find the gradient of the line that goes through the points:

1
$(1, 4)(3, 10)$

2
$(5, 7)(8, 1)$

3
$(2, 2)(-1, 10)$

4
Show that the points $\mathrm{A}(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are ‘collinear’ they lie on the same line.

Chapter 5 · Original slide 10 · Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1.
Work out the value of a.

Chapter 5 · Original slide 11 · y=mx+c

One form we can put a straight line equation in is:

$y = mx + c$

Gradient

y-intercept

Why does it work?

• The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$. Substituting:
$\mathbf{y=(m\times 0)+c=c}$ as expected.

• By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:
$\mathbf{y=m\,(x+1)+c=mx+m+c}$
which indeed has increased by $\mathbf{m}$.

Chapter 5 · Original slide 12 · y=mx+c

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form

$y = mx + c$

Put y on the side it's positive.

Divide each term by 3; don't write $y = \frac{4x+5}{3}$ otherwise it's not in the form $y = mx + c$

This is algebra, so use improper fractions, and not mixed numbers or recurring decimals.

Chapter 5 · Original slide 13 · ax+by+c=0

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the 'slope-intercept form'.

But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers. This is known as the 'standard' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation. $-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

Chapter 5 · Original slide 14 · Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$2x + y = 4$



$y = mx + c$

“Slope-Intercept Form”

$ax + by + c = 0$

“Standard Form”

Coverage

$y = mx + c$ doesn’t allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

Usefulness

This more ‘elegant’ form also means it ties in with vectors and matrices. In FM, you will learn about the ‘dot product’ of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 (1)$$

Symmetry

In general, the ‘linear combination’ of two variables x and y is $ax + by$, i.e. “some amount of x and some amount of y”. There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

We can extend to 3D points to get the equation of a plane:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is perpendicular to the line. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$x + 4 = 0$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$$

$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Chapter 5 · Original slide 15 · Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

Chapter 5 · Original slide 16 · Exercise 5A – Pure Y1 Page 91

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. (3 marks)

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

Chapter 5 · Original slide 17 · Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

Chapter 5 · Original slide 18 · Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

Chapter 9 · Original slide 44 · Sin Graph

90 180 270 360

What does it look like?

Key Features:

• Repeats every $360°$.

• Range: $-1 < \sin(x) < 1$

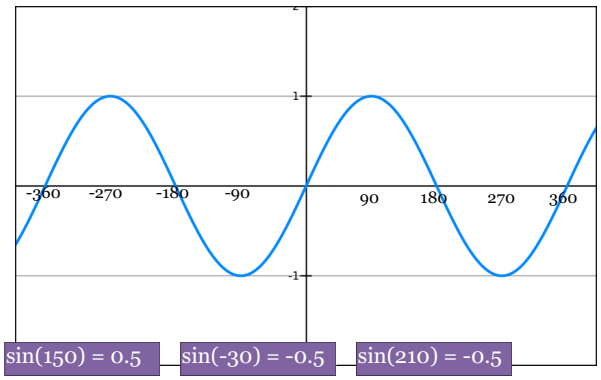
-360

-270

-180

-90

Chapter 9 · Original slide 45 · Sin Graph



What do the following graphs look like?

Suppose we know that sin(30) = 0.5. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 46 · Cos Graph

90 180 270 360

What do the following graphs look like?

Key Features:

- Repeats every 360°.
- Range: $-1 < \sin(x) < 1$

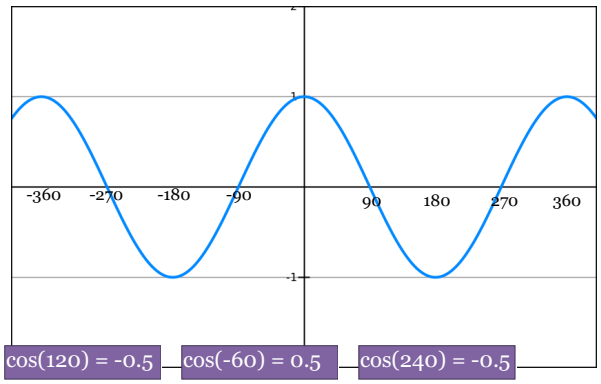
-360

-270

-180

-90

Chapter 9 · Original slide 47 · Cos Graph



What does it look like?

Suppose we know that cos(60) = 0.5. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 48 · Tan Graph

90 180 270 360

What does it look like?

-360

-270

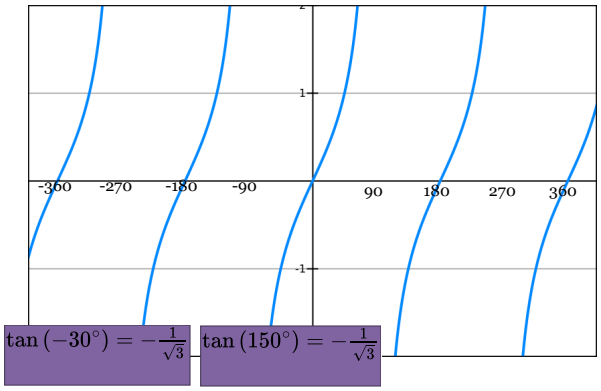
-180

-90

Key Features:

- Repeats every $\mathbf{180}^\circ$.
- Roots: $0, 180^\circ, -180^\circ, \ldots$
- Range: $\tan(x) \in \mathbb{R}$ (i.e. no min/max value!)
- Asymptotes: $x = \pm 90^\circ, \pm 270^\circ, \ldots$

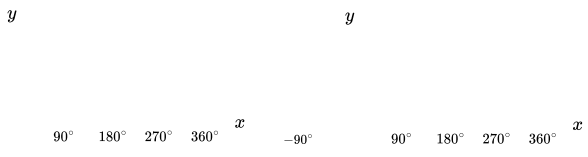
Chapter 9 · Original slide 49 · Tan Graph



What does it look like?

Suppose we know that $\tan(30°) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

Chapter 9 · Original slide 50 · Transforming Trigonometric Graphs



There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs 'inside' the function (i.e. input modified) or 'outside' the function (i.e. output modified).

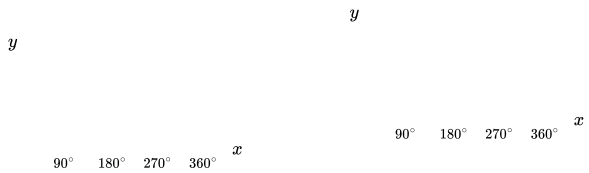
Sketch $y = 4\sin x$, $0 \leq x \leq 360°$

Sketch $y = \cos(x + 45°)$, $0 \leq x \leq 360°$

Graphs and Transformations

Chapter 9 · Original slide 51 · Transforming Trigonometric Graphs

Sketch $y = -\tan x, 0 \leq x \leq 360^\circ$



Sketch $y = \sin\left(\frac{x}{2}\right), 0 \leq x \leq 360^\circ$

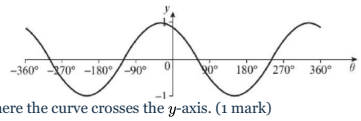
Chapter 9 · Original slide 52 · Exercise 9F – Y1 Page 194

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.
3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

Chapter 9 · Original slide 53 · Exercise 9G – Y1 Pages 197-198

The graph shows the curve $y = \cos(x + 30°)$, $-360° \leqslant x \leqslant 360°$.

a Write down the coordinates of the points where the curve crosses the x-axis. **(2 marks)**

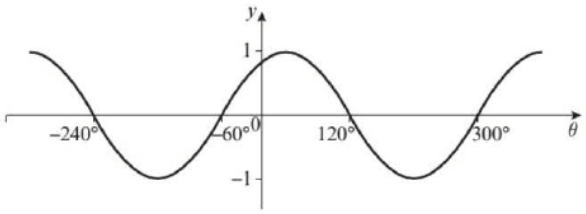


b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

7 The graph shows the curve $y = \cos(x + 30°)$, $-360° \leq x \leq 360°$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)

b $\cos 30° = \dfrac{\sqrt{3}}{2}$; $\left(0, \dfrac{\sqrt{3}}{2}\right)$

Chapter 9 · Original slide 54 · Exercise 9G – Y1 Pages 197-198

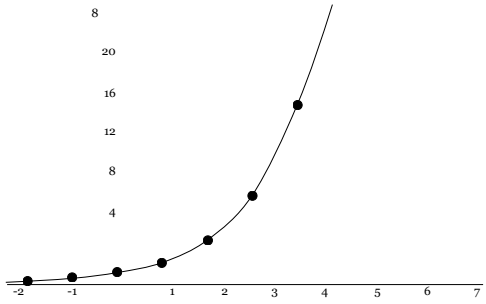


8 The graph shows the curve with equation $y = \sin(x + k)$, $-360° \leq x \leq 360°$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)

Chapter 9 · Original slide 55 · Exercise 9G – Y1 Pages 197-198

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Chapter 14 · Original slide 4 · y={2}^{x}



x	-2	-1	0	1	2	3	4
y							

Note: Ensure that you can distinguish between a x^a (e.g. polynomial) term and an a^x exponential term. In the former the variable is in the base, and in the letter the variable is in the power. 2^x behaves very differently to x^2, both in its rate of growth (i.e. exponential terms grow much faster!) and how it differentiates.

Why are exponential functions important?

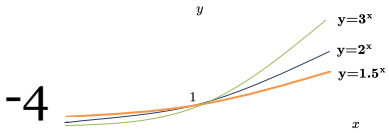
Each of the common graphs have a key property that makes them useful for modelling.

For reciprocal graphs $y = \frac{k}{x}$, as x doubles, y halves. This means we'd use it to represent variables which are inversely proportional.

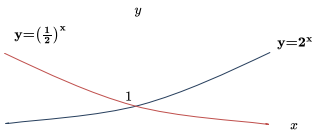
Linear graphs are used when we're adding the same amount each time.

In contrast, exponential graphs are used when we're multiplying by the same amount each time. For example, we might use $1000(1.05^t)$ to model our savings with

Chapter 14 · Original slide 5 · Contrasting exponential graphs



On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$



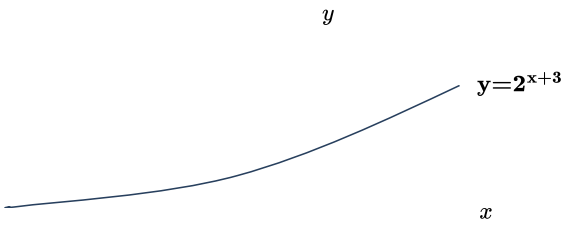
On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

Three important notes:

- $y = 2^x$ is said to be "exponential growing" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "exponentially decaying", because it's getting smaller (halving) each time x increases by 1.
- $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$. Proof: If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$
- $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.

You should therefore in general be able to recognise and sketch the graph $\mathbf{y=a^{-x}}$.

Chapter 14 · Original slide 6 · Graph Transformations



Sketch $y = 2^{x+3}$

The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

Chapter 14 · Original slide 7 · Exercise 14A (Pages 313-314)

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

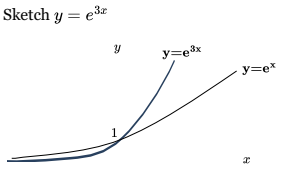
7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

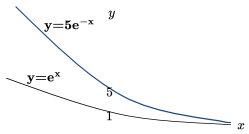
b Find the values of the constants p and q.

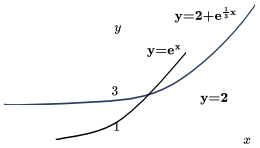
Chapter 14 · Original slide 11 · More Graph Transformations

Sketch $y = e^{3x}$



Sketch $y = 5e^{-x}$





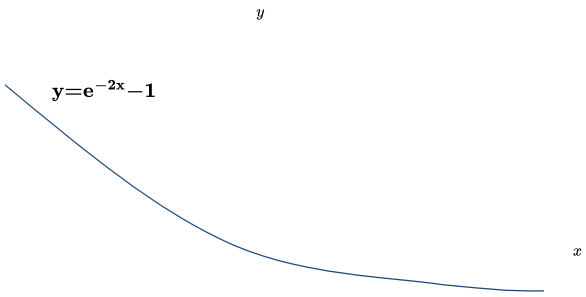
Note: Recall the shape of a $y = a^{-x}$ graph.

5 is 'outside' function, so stretch of scale factor 5 on y-axis.

Important Note: Because the original asymptote was $y = 0$, it is now $y = 2$ and you must indicate this along with its equation.

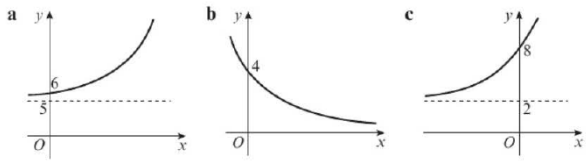
Sketch $y = 2 + e^{\frac{1}{3}x}$

Chapter 14 · Original slide 12 · Test Your Understanding



Sketch $y = e^{-2x} - 1$

Chapter 14 · Original slide 13 · Exercise 14B (Pages 316-317)



4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.

Chapter 14 · Original slide 14 · Exercise 14B (Pages 316-317)

6 Differentiate the following with respect to x.

a e^{6x} **b** $e^{-x/3}$ **c** $7e^{2x}$ **d** $5e^{0.4x}$ **e** $e^{3x} + 2e^{x}$ **f** $e^{x}(e^{x} + 1)$

Chapter 14 · Original slide 15 · Exercise 14B (Pages 316-317)

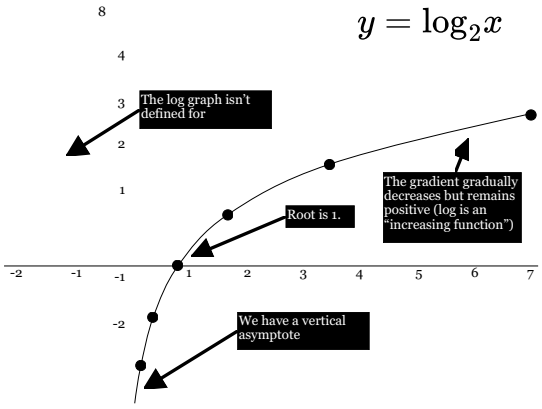
7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

a $x = 2$ **b** $x = 0$ **c** $x = -0.5$

8 The function f is defined as $\mathrm{f}(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point $(5, e)$ goes through the origin.

Graphs and Transformations

Chapter 14 · Original slide 27 · Exponentials and logarithms



x	0.25	0.5	1	2	4	8
y						

Chapter 14 · Original slide 41 · Natural Logarithms

We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.

We also saw that e^x is "the" exponential function.

The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

! The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x =$

$e^{\ln x} =$

Since "e to the power of" and "ln of" are inverse functions, they cancel each other out!

Solve $e^x = 5$

"ln both sides".

On the LHS it cancels out the "e to the power of"

Solve $2 \ln x + 1 = 5$

Do "e to the power of" each side. On the LHS it cancels out the ln.